The Sticky Kakeya Conjecture

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Contents

1 Introduction **3**
1.1 Motivation 3
1.2 Outline of the report 3

2 Preliminaries **4**
2.1 Kakeya Sets 4
2.2 The Besicovitch Construction (Perron Tree) 4
2.3 Minkowski and Hausdorff Dimensions 5
2.4 The Kakeya Conjecture 7

3 Sticky Kakeya Sets **7**
3.1 Definitions 8
3.2 The Worst Counterexample Framework 10

4 Structural Properties of Sticky Sets **11**
4.1 Coarse Self-Similarity 11
4.2 Plaininess and Graininess 14

5 Proof of the Sticky Kakeya Conjecture **17**

6 Conclusion **20**
6.1 Acknowledgements 21

1 Introduction

1.1 Motivation

In 1917, Soichi Kakeya posed an elementary question: what is the region of smallest area in which a unit-length needle can be continously rotated through a full 180° and returned to its starting position? The surprising answer, due to Besicovitch (1919) [1], is that there is no smallest area — the needle can be rotated within a region of arbitrarily small area. The construction achieving this reveals something stranger than the original question intended: it produces a compact set of Lebesgue measure zero that nonetheless contains a full unit line segment pointing in every direction in the plane.

Once such sets — hereon called Kakeya sets — are known to have zero area, the notion of area and volume are no longer useful for distinguishing "large" Kakeya sets from "small" ones; a finer notion is needed. Hausdorff dimension supplies exactly this. The Kakeya conjecture is the assertion that despite having measure zero, every Kakeya set in $\mathbb{R}^n$ is as large as it possibly could be in this dimensional sense: it has full Hausdorff dimension n.

This conjecture, while phrased as an elementary geometric statement, sits at the center of a much larger web of problems in modern analysis. A Kakeya-type estimate is a necessary ingredient for Stein's restriction conjecture in Fourier analysis, which governs how the Fourier transform of a function can be meaningfully restricted to a curved surface; it is closely tied to the Bochner-Riesz conjecture on the convergence of Fourier series and integrals; it controls local smoothing estimated for solutions of the wave equation; and closely related combinatorial incidence bounds — via the "polynomial method" following Dvir's resolution of the finite-field analogue and Guth and Katz's resolution of the joints problem — have become a standard tool throughout incidence geometry and analytic number theory.

The Kakeya conjecture is thus not merely a curiosity about rotating needles, but a genuine bottleneck problem whose resolution (even in special cases) tends to produce techniques with far broader reach. This report focuses on the recent proof of the conjecture in $\mathbb{R}^3$ by Hong Wang and Joshua Zahl [2], and in particular the proof for the special case of "sticky" Kakeya sets in $\mathbb{R}^3$.

1.2 Outline of the report

Section 2 develops the necessary preliminaries: the formal definition of a Kakeya set, a sketch of the Besicovitch construction showing such sets can have meausure zero, the definitions of Hausdorff and Minkowski dimension, and the formal statement of the Kakeya conjecture. Section 3 introduces the sticky Kakeya sets and the "worst counterexample" framework that reduces the sticky Kakeya conjecture to a single extremal quantity, σ.

Section 5 establishes the three structural properties — coarse self-similarity,

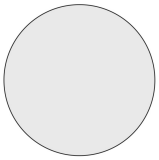
plaininess, and graininess — that any extremal sticky configuration must possess. Section 6 gives the proof that $\sigma = 0$ assuming Wolff's $L^{3/2}$ circular maximal function bound for C^2 curves.

2 Preliminaries

2.1 Kakeya Sets

Definition 2.1 (Kakeya Set)**.** A compact set $K \subseteq \mathbb{R}^n$ is Kakeya if, for every unit vector $e \in S^{n-1}$, K contains a unit line segment parallel to e.

Example. Closed unit disk in $\mathbb{R}^2$ or the closed unit sphere in $\mathbb{R}^3$.

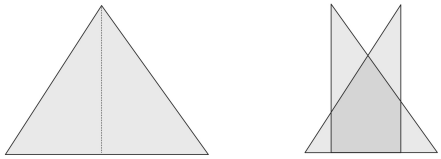


A disk is, however, far from the most economical Kakeya set in terms of area. A single equilateral triangle in $\mathbb{R}^2$ is sufficient to cover one-third of the possible directions $e \in S^1$ be placing one end of the needle on the top vertex and the other end on the bottom edge. Another example of smaller area is the deltoid.

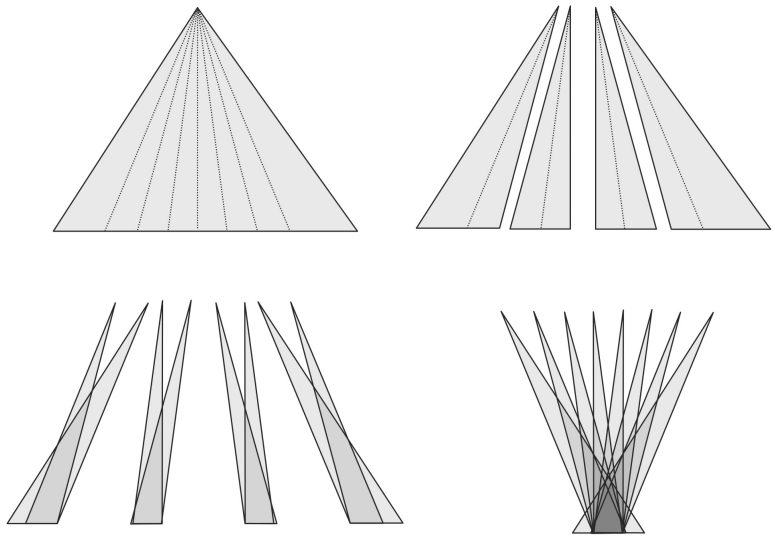
Perhaps, the most instructive example is the triangle which was the idea behind Besicovitch's construction; further refined by Perron.

2.2 The Besicovitch Construction (Perron Tree)

The key idea is a splitting-and-sliding trick on the triangle that preserves the amount of directions covered by the top vertex, while reducing the area significantly. The following pictures sketch the idea.



The resulting figure on the right has nearly half the area of the initial triangle, while the top two vertices still cover as much directions. In general we can split the base into 2^n segments, and the same technique would give us a shape which has area $\frac{1}{2^n}$ times the area of the original triangle.



By having 3 copies of these "Perron Trees" each rotated by 60°, we can have Kakeya sets of arbitrarily small area. A limiting arugment on these constructions was used by Besicovitch [1] to show the following.

Theorem 2.2 (Besicovitch, 1919)**.** For every $n \geq 2$, there exists Kakeya sets $K \subseteq \mathbb{R}^n$ of Lebesgue measure zero.

Remark. The above theorem showed that Lebesgue measure cannot be used to capture the "largeness" of Kakeya sets, and a finer invariant was required. Kakeya conjectured that perhaps Hausdorff dimension is the right idea.

2.3 Minkowski and Hausdorff Dimensions

Let $K \subseteq \mathbb{R}^n$ be a totally bounded set. By $N(K, \varepsilon)$, denote the minimal number of sets of diameter less than ε required to cover K.

Definition 2.3 (Minkowski Dimension)**.** The upper and lower Minkowski Dimensions are defined as

$$\overline{\dim_{\mathcal{M}}}(K) := \limsup_{\varepsilon \to 0} \frac{\log(N(K,\varepsilon))}{\log(1/\varepsilon)} \qquad \underline{\dim_{\mathcal{M}}}(K) := \liminf_{\varepsilon \to 0} \frac{\log(N(K,\varepsilon))}{\log(1/\varepsilon)}$$

If these two limits agree, then it is called the Minkowski Dimension of K and it is denoted by $\dim_{\mathcal{M}}(K)$.

Let $N_B(K,\varepsilon)$ denote the minimal number of open balls of radius ε required to cover K. It is not difficult to see that $N_B(K,\varepsilon) \leq N(K,\varepsilon) \leq N_B(K,\varepsilon/2)$. Thus, we can safely replace N with N_B in **Definition 2.3**.

Definition 2.4 (ε-separatedness)**.** (X,d) be a metric space and $A \subseteq X$. For $\varepsilon > 0$, A is called ε-separated if $(\forall x, y \in A)(x \neq y \Rightarrow d(x,y) \geq \varepsilon)$.

Let $N_{\text{sep}}(K,\varepsilon)$ denote the cardinality of a maximal ε-separated subset of K. It is again not difficult to see that $N(K,\varepsilon) \leq N_{\text{sep}}(K,\varepsilon) \leq N(K,\varepsilon/2)$.

While considering Minkowski Dimensions of $K \subseteq \mathbb{R}^n$, without loss of generality, we can consider either covers of K formed by open balls of radius ε or maximal ε-separated subsets of K.

For more on Minkowski dimension, we refer the reader to Bishop [3].

Definition 2.5 (α-dimensional Hausdorff Content)**.** Let $K \subseteq \mathbb{R}^n$. Define

$$\mathcal{H}^\alpha_\infty(K) := \inf\left\{\sum_i \operatorname{diam}(U_i)^\alpha : K \subseteq \bigcup_i U_i\right\}$$

where infimum is taken over $\{U_i\}$'s, countable covers of K using any sets.

Suppose $\mathcal{H}^\alpha_\infty(K) = 0$ for some α and $\beta > \alpha$. Then for every $\varepsilon > 0$, there is a countable cover $\{U_i\}$ of K such that

$$\sum_i \operatorname{diam}(U_i)^\alpha < \varepsilon \Rightarrow \sum_i \operatorname{diam}(U_i)^\beta = \sum_i \operatorname{diam}(U_i)^\alpha \cdot \operatorname{diam}(U_i)^{\beta-\alpha}$$

If K is bounded, we can assume $\operatorname{diam}(U_i) \leq \operatorname{diam}(K) = D$ for every i. So, U_i forms a cover whose β-content is less than $D^{\beta-\alpha}\varepsilon$. This shows that content is monotone in α and in particular $\mathcal{H}^\beta_\infty(K) = 0$ by letting $\varepsilon \to 0$.

Definition 2.6 (Hausdorff Dimension)**.** $K \subseteq \mathbb{R}^n$ be bounded.

$$\dim_{\mathcal{H}}(K) := \inf\{\alpha : \mathcal{H}^\alpha_\infty(K) = 0\}$$

The following is an important relationship between Hausdorff and Minkowski dimensions.

Proposition 2.7. $\dim_{\mathcal{H}}(K) \leq \underline{\dim_{\mathcal{M}}}(K) \leq \overline{\dim_{\mathcal{M}}}(K)$

Proof. Let $B_i = B(x_i, \varepsilon/2)$ be $N(K,\varepsilon)$-many balls of radius $\varepsilon/2$ with centers $x_i \in K$ that cover K. Then, noting that $\operatorname{diam}(B_i) = \varepsilon$,

$$S_\varepsilon = \sum_{i=1}^{N(K,\varepsilon)} \varepsilon^\alpha = N(K,\varepsilon)\varepsilon^\alpha = \varepsilon^{\alpha - R_\varepsilon}$$

where $R_\varepsilon = \frac{\log N(K,\varepsilon)}{\log 1/\varepsilon}$. Now suppose $\alpha > \underline{\dim}_{\mathcal{M}}(K) = \liminf_{\varepsilon \to 0} R_\varepsilon$. So there is a subsequence ε_k which goes to 0 as $k \to \infty$ and $\alpha - R_{\varepsilon_k} > 0$ for all k. So, $\inf_{\varepsilon_k} S_{\varepsilon_k} = 0$ and in particular $\inf_{\varepsilon > 0} S_\varepsilon = 0$. If follows that $\dim_{\mathcal{H}}(K) \leq \underline{\dim}_{\mathcal{M}}(K)$. The second inequality follows from definition □

For more on Hausdorff dimension, we refer the reader to Bishop [3] and for more on Hausdorff meausre, we refer the reader to Folland [4].

2.4 The Kakeya Conjecture

Conjecture 2.8 (Kakeya)**.** If $K \subseteq \mathbb{R}^n$ is a Kakeya set, then $\dim_{\mathcal{H}}(K) = n$.

Since $K \subseteq \mathbb{R}^n$, we trivially have that $\dim_{\mathcal{H}}(K) \leq \dim_{\mathcal{H}}(\mathbb{R}^n) = n$. The difficulty of the conjecture is pushing the lower bound all the way up to n, for a potentially measure zero set.

The conjecture is completely resolved only in low dimensions: it is trivial for $n = 1$; it was proved for $n = 2$ by Davies [5] in 1971; and as of 2025, it has been proved for $n = 3$ by Wang & Zahl [6]. The proof of the $n = 3$ case relies on Wang & Zahl's previous work [2] on the special class of Kakeya sets, namely the sticky Kakeya sets, which is the focus of the remainder of this report. For $n \geq 4$, the conjecture remains open, though some non-trivial partial lower bounds are known.

Theorem 2.9 (Wang & Zahl, 2025)**.** If $K \subseteq \mathbb{R}^3$ is a Kakeya set, then $\dim_{\mathcal{H}}(K) = 3$.

3 Sticky Kakeya Sets

The class of all Kakeya sets in $\mathbb{R}^3$ is extremely unstructured. While **Theorem 2.2** shows that they can have measure zero, it places essentially no constraint on their shape. A standard and powerful strategy is to identify a class of Kakeya sets which have useful structural properties and separately show that the general case can be reduced to (or dominated by) this special class.

Sticky Kakeya sets are exactly such a special class. They were isolated by Katz, Laba, and Tao [7]. The sticky configurations emerged as a natural candidate for Kakeya sets with Hausdorff dimension strictly less than n. Intuitively, a hypothetical counterexample to the Kakeya set would have to be "sticky".

3.1 Definitions

Definition 3.1 (Sticky Kakeya Sets)**.** A compact set $K \subseteq \mathbb{R}^n$ is a sticky Kakeya set, if there exists a set of lines L with packing dimension $n-1$ that contains atleast one line in every direction and for each $l \in L$, $l \cap K$ contains a unit interval.

We shall now formally define the standard tool called δ-tubes and state the result that reduces sticky Kakeya sets to an extremal quantity of these δ-tubes.

$$\mathcal{L} := \left\{ (p, v) : p \in \left[\frac{-1}{3}, \frac{1}{3}\right]^2, v \in S^2 \text{ with } v_3 \geq \frac{1}{2} \right\}$$

By $l = (p, v) \in \mathcal{L}$ we represent the line $(p, 0) + \mathbb{R}v \in \mathbb{R}^3$. Note that $\mathcal{L}$ inherits a metric and a measure from the parent spaces. In particular, for $l_1, l_2 \in \mathcal{L}$ where $l_i = (p_i, v_i)$ for $i = 1, 2$ we have $d(l_1, l_2) = \|p_1 - p_2\| + \|v_1 - v_2\|$ and the measure is the product measure $\lambda := \mu \times \nu$, where μ is the standard Lebesgue measure and ν is the surface measure on S^2.

Definition 3.2 (δ-tube)**.** For $\delta > 0$ and $l \in \mathcal{L}$, the δ-tube of l is defined as

$$T := N_{6\delta}(l) \cap [-1, 1]^3$$

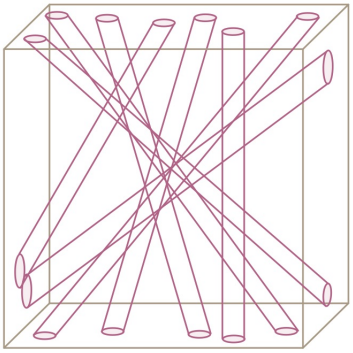
If $C > 0$, then CT denotes the $C\delta$-tube of l.

Definition 3.3 (δ-cube)**.** For $\delta > 0$, a δ-cube is a set of the form $p + [0, \delta)^n$ where $p \in (\delta\mathbb{Z})^n$.

We use the $2n\delta$-neighborhood (with $n = 3$) in the definition of a δ-tube rather than the δ-neighborhood so that any δ-cube intersecting l will be contained in the δ-tube of l.

Definition 3.4 (Shading)**.** Given a δ-tube T, a shading of T is a union of δ-cubes $Y(T)$ such that $Y(T) \subseteq T$.

By $(\mathbb{T}, Y)_\delta$, we will denote a collection of δ-tubes and their associated shadings. The following picture is how a typical collection looks.



For $T, T' \in \mathbb{T}$, we say

- They are *essentially identical* if $d(l, l') \leq \delta$; otherwise *essentially distinct*.
- They are *essentially parallel* if $\|v - v'\| \leq \delta$.

Definition 3.5 (Subcollections)**.** $(\mathbb{T}', Y')_\delta$ is a subcollection of $(\mathbb{T}, Y)_\delta$ if $\mathbb{T}' \subseteq \mathbb{T}$ and for all tubes $T' \in \mathbb{T}'$, we have $Y'(T') \subseteq Y(T')$.

In this section we will show that for finite collections of tubes, we can always move to a sufficiently good collection without loss of generality.

Notation. Given $(\mathbb{T}, Y)_\delta$, denote $E_{\mathbb{T}} := \bigcup_{T \in \mathbb{T}} Y(T)$.

Definition 3.6 (Refinement)**.** A subcollection is called a refinement if

$$\sum_{T' \in \mathbb{T}'} |Y'(T')| \geq \frac{1}{\left(\log\left(\frac{1}{\delta}\right)\right)^C} \sum_{T \in \mathbb{T}} |Y(T)| \quad \text{where } C \geq 0 \text{ is some constant}$$

We will denote this by $\sum_{T' \in \mathbb{T}'} |Y'(T')| \gtrapprox_\delta \sum_{T \in \mathbb{T}} |Y(T)|$.

A refinement $(\mathbb{T}', Y')_\delta$ is said to have *constant multiplicity* if the following holds.

$$\sum_{T \in \mathbb{T}'} \chi_{Y'(T)} = M \cdot \chi_{E_{\mathbb{T}'}} \quad \text{for some integer } M \geq 1$$

For $\rho \geq \delta > 0$, a ρ-tube $\tilde{T}$ is said to *cover* a δ-tube T if $d(l, \tilde{l}) \leq \rho/2$. Along the same lines, a collection $\tilde{\mathbb{T}}$ is said to cover another collection $\mathbb{T}$ if

$$(\forall T \in \mathbb{T})(\exists \tilde{T} \in \tilde{\mathbb{T}})(\tilde{T} \text{ covers } T)$$

Notation. Denote $\mathbb{T}[\tilde{T}] := \{T \in \mathbb{T} : \tilde{T} \text{ covers } T\}$

Furthermore, $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ covers $(\mathbb{T}, Y)_\delta$ if in addition to the above, we have

$$(\forall \tilde{T} \in \tilde{\mathbb{T}})(\forall T \in \mathbb{T}[\tilde{T}])(Y(T) \subseteq \tilde{Y}(\tilde{T}))$$

Such a covering is called *balanced* if $|E_{\mathbb{T}} \cap Q|$ is the same for each ρ-cube $Q \subseteq E_{\tilde{\mathbb{T}}}$.

3.2 The Worst Counterexample Framework

For $s, t, \delta \in (0, 1)$, define

$$M(s, t, \delta) = \inf_{(\mathbb{T}, Y)_\delta} |E_{\mathbb{T}}|$$

where the infimum is taken over $(\mathbb{T}, Y)_\delta$ satisfying:

(a) Tubes in $\mathbb{T}$ are essentially distinct.

(b) For every $\rho \in [\delta, 1]$, $(\mathbb{T}, Y)_\delta$ has a ρ-cover that has atmost δ^{-t}-many mutually essentially parallel ρ-tubes.

(c) $\sum_{T \in \mathbb{T}} |Y(T)| \geq \delta^s$.

Remark. Taking $\rho = \delta$, in condition (b) shows that as $t \to 0$, the tubes in $(\mathbb{T}, Y)_\delta$ are forced to be δ-separated.

Next, define

$$N(s, t) = \limsup_{\delta \to 0^+} \frac{\log M(s, t, \delta)}{\log \delta}$$

Note that $N(s, t)$ is jointly decreasing in both s and t and also bounded, $0 \leq N(s, t) \leq n$. Thus,

$$\lim_{(s,t) \to (0,0)} N(s, t) = \inf_{s,t \in (0,1)} N(s, t) =: \sigma$$

The main result that helps us move from sticky Kakeya sets to δ-tubes is as follows.

Proposition 3.7. If $K \subseteq \mathbb{R}^3$ is a sticky Kakeya set, then $\dim_{\mathcal{H}}(K) \geq 3 - \sigma$.

We do not give the proof of the above proposition here, and instead refer the reader to Wang & Zahl [2]. There is no reason as to why σ should be achieved. However, since σ is a limit at $(s, t) = (0, 0)$ it is useful to look at collection $(\mathbb{T}, Y)_\delta$ for which (a), (b) and (c) hold with $s = t = \varepsilon$ and analyze their properties as $\varepsilon \to 0$. To this end, we define

Definition 3.8 (Extremal Families)**.** Let $\varepsilon, \delta > 0$. A collection $(\mathbb{T}, Y)_\delta$ is said to be ε-extremal if it satisfies (a), (b) and (c) with $s = t = \varepsilon$ and furthermore,

$$|E_{\mathbb{T}}| \leq \delta^{\sigma - \varepsilon}$$

Observe. For small δ, extremal families always exist.

Proof. Given $\varepsilon, \delta_0 > 0$, we will show that $\exists \delta < \delta_0$ such that there is an ε-extremal family $(\mathbb{T}, Y)_\delta$. Towards a contradiction suppose not. The only this can happen is if for all collections $(\mathbb{T}, Y)_\delta$ with $\delta < \delta_0$ for which (a), (b) and (c) hold, $|E_\mathbb{T}| > \delta^{\sigma-\varepsilon}$. This implies

$$M(\varepsilon, \varepsilon, \delta) \geq \delta^{\sigma-\varepsilon} \Rightarrow N(\varepsilon, \varepsilon) \leq \sigma - \varepsilon < \sigma$$

Since N is decreasing in both coordinates, $\sigma \leq N(\varepsilon, \varepsilon) < \sigma$. Contradiction. □

In the next section, we will look at the structural properties of extremal families.

4 Structural Properties of Sticky Sets

A definition is in order, before we can state the main results.

Definition 4.1 (Unit Rescaling of a Tube)**.**

The proofs of the results in this section are beyond the scope of the report and we refer the reader to Wang & Zahl [**?**].

4.1 Coarse Self-Similarity

Proposition 4.2. $(\forall \varepsilon > 0)(\exists \eta, \delta_0 > 0)(\forall \delta \in (0, \delta_0])$, we have:

IF: $(\mathbb{T}, Y)_\delta$ is η-extremal and $\rho \in [\delta^{1-\varepsilon}, \delta^\varepsilon]$

THEN: $(\mathbb{T}, Y)_\delta$ has a refinement $(\mathbb{T}', Y')_\delta$ and a balanced cover $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$,

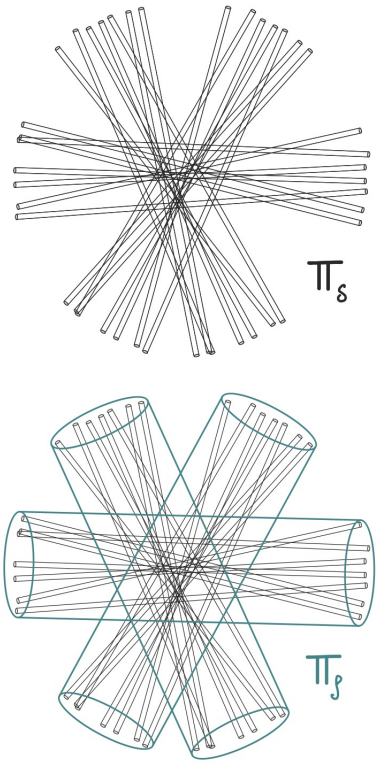
(i) $(\tilde{\mathbb{T}}, \tilde{Y})_\rho$ is ε-extremal.

(ii) For every $\tilde{T} \in \tilde{\mathbb{T}}$, the unit rescaling of $(\mathbb{T}, Y)_\delta[\tilde{T}]$ is ε-extremal.

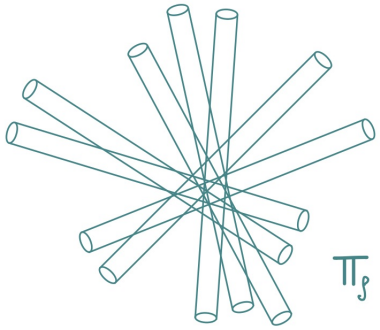
(iii) For each $p \in \mathbb{R}^3$, $\#\tilde{\mathbb{T}}(p) \leq \rho^{-\sigma-\varepsilon}$.

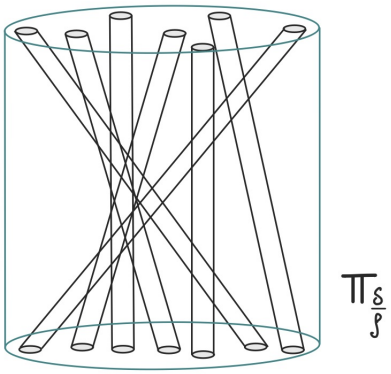
(iv) For each $p \in \mathbb{R}^3$ and $\tilde{T} \in \tilde{\mathbb{T}}$, $\#\mathbb{T}[\tilde{T}](p) \leq \left(\frac{\delta}{\rho}\right)^{-\sigma-\varepsilon}$

We will try to give some intuition behind this result. For small δ, proposition — is saying that there are extremal collections which are self-similar at particular large/coarse scales. In particular, if the following (above) represents the refinement of any η-extremal collection, then we have a balanced cover (below).



Furthermore, both the unit rescaling with respect to a single covering tube (above) as well as the entire covering collection (below) are ε-extremal.





This is a very strong form of self-similarity that is occuring at many different scales at the same time. A familiar example of such a phenomenon is the Cantor set which exhibits extreme self-similarity at all scales.

Some motivation for Ahlfors-David regularity:

Example (The Hausdorff dimension of the Cantor set)**.**

Recall the construction of the Cantor set as $K = \cap_n K_n$. The 2^n-many intervals of diameter 3^{-n} in K_n form a cover of K. Thus, taking $\alpha_0 = \frac{\log 2}{\log 3}$,

$$\sum_{i=1}^{2^n} (3^{-n})^{\alpha} = 2^n (3^{-n})^{\alpha} = 3^{(\alpha_0 - \alpha)n}$$

So, if $\alpha > \alpha_0$, taking $n \to \infty$ sends the above to 0. Thus, $\mathcal{H}^{\alpha}_{\infty}(K) = 0$ for $\alpha > \alpha_0$ which implies $\dim_{\mathcal{H}}(K) \leq \alpha_0$. The other direction requires the Mass Distribution Principle and we refer the reader to Bishop [3].

Remark. The self-similarity of the Cantor set ensures that for all n, we have $N(K, 3^{-n}) = 2^n$. This is essentially what Ahlfors-David regularity is trying to capture. More precisely, we need a weaker form of Ahlfors-David regularity which we define below.

Definition 4.3 (ADset)**.** Let $\alpha \in (0, 1], C \geq 1, \delta > 0$. A set $E \subseteq \mathbb{R}$ is called an (δ, α, C)-ADset if $\forall x \in \mathbb{R}$

$$N(E \cap B(x, r), \rho) \leq C(r/\rho)^{\alpha} \quad \text{for all } \rho \geq \delta \text{ and } r \geq \rho$$

In particular, we can take $\alpha = 1 - \sigma$, $r = 1$, $C = \delta^{-\varepsilon}$ and $\rho = \delta$. Furthermore, assuming $E \subseteq B(x, r)$ for some $x \in \mathbb{R}$, then

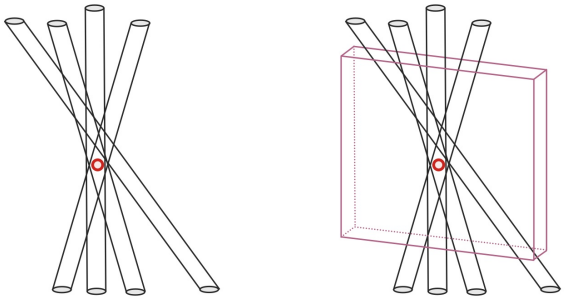
$$N(E, \delta) \leq \delta^{-\varepsilon} \delta^{\sigma - 1}$$

which means, there is a cover of E using atmost $\delta^{\sigma - 1 - \varepsilon}$ intervals of length δ. The choice of these parameters are made in retrospect, and will be useful later.

4.2 Plaininess and Graininess

Definition 4.4 (Plane Map)**.** A plane map, given a set of δ-tubes $(\mathbb{T}, Y)_\delta$, is a function $V : E_{\mathbb{T}} \to S^2$ satisfying

$$|\mathrm{dir}(T) \cdot V(p)| \leq \delta \quad \text{for all } p \in E_{\mathbb{T}}, T \in \mathbb{T}$$



Essentially, the existence of a plane map says that the subcollection of δ-tubes passing through a fixed point are roughly coplanar, upto resolution δ. In other words, these tubes are contained inside slabs of width $\sim \delta$ as shown above.

So, if we are given the existence of a plane map, the only other missing piece of information is the gap between these slabs and the rate at which the orientation (or normal direction) of the slabs change as we change the point $p \in E_{\mathbb{T}}$.

This and much more are given by the next result.

Proposition 4.5. For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and an ε-extremal family $(\mathbb{T}, Y)_\delta$ with the following properties.

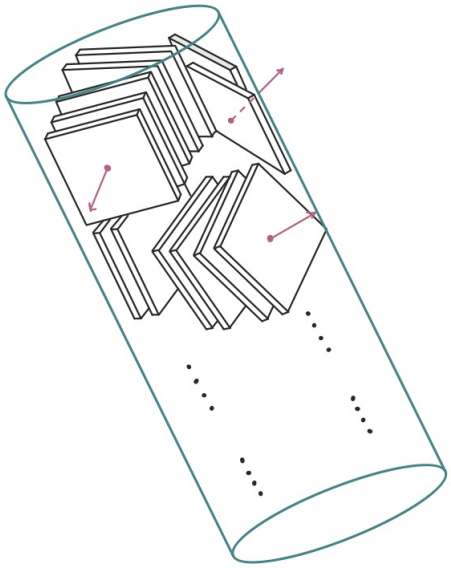
1. $E_{\mathbb{T}}$ **is a union of local grains with a Lipschitz plane map**:

   There is a 1-Lipschitz plane map $V : E_{\mathbb{T}} \to S^2$, such that for all $p \in \mathbb{R}^3, \rho \in [\delta, 1]$, $V(p) \cdot (B(p, \sqrt{\rho}) \cap E_{\mathbb{T}})$ is an $(\rho, 1 - \sigma, \delta^{-\varepsilon})$-ADset.

2. $E_{\mathbb{T}}$ **is a union of global grains with a Lipschitz slope function**:

   There is a 1-Lipschitz slope function $f : [-1, 1] \to \mathbb{R}$, such that for all $z \in [-1, 1]$, $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is an $(\delta, 1 - \sigma, \delta^{-\varepsilon})$-ADset.

Remark. The first part of proposition — essentially forces the tubes to behave similar to the picture below.



Each local neighbourhood $B(p, \sqrt{\rho}) \cap E_{\mathbb{T}}$ which is essentially a cluster of δ-cubes, when projected along $V(p)$ gives us an ADset, say F, satisfying

$$N(F, \rho) \leq \delta^{-\varepsilon} \left(\frac{\sqrt{\rho}}{\rho} \right)^{1-\sigma} = \delta^{-\varepsilon} \rho^{\sigma/2 - 1/2}$$

So, F can be covered by atmost $\delta^{-\varepsilon}\rho^{\sigma/2-1/2}$-many intervals of length ρ. If we were to extend these covering intervals in the direction orthogonal to $V(p)$, we would get slabs with dimensions of the order $\sqrt{\rho} \times \sqrt{\rho} \times \rho$ which would cover $B(p, \sqrt{\rho})$. So, locally, the set $E_{\mathbb{T}}$ is a union of slabs (called local grains).

Furthermore, recall that being ε-extremal, $|E_{\mathbb{T}}| \leq \delta^{\sigma-\varepsilon}$. Thus, $E_{\mathbb{T}}$ can be covered by $\delta^{\sigma-\varepsilon}\rho^{-3/2}$-many balls of radius $\sqrt{\rho}$. So roughly,

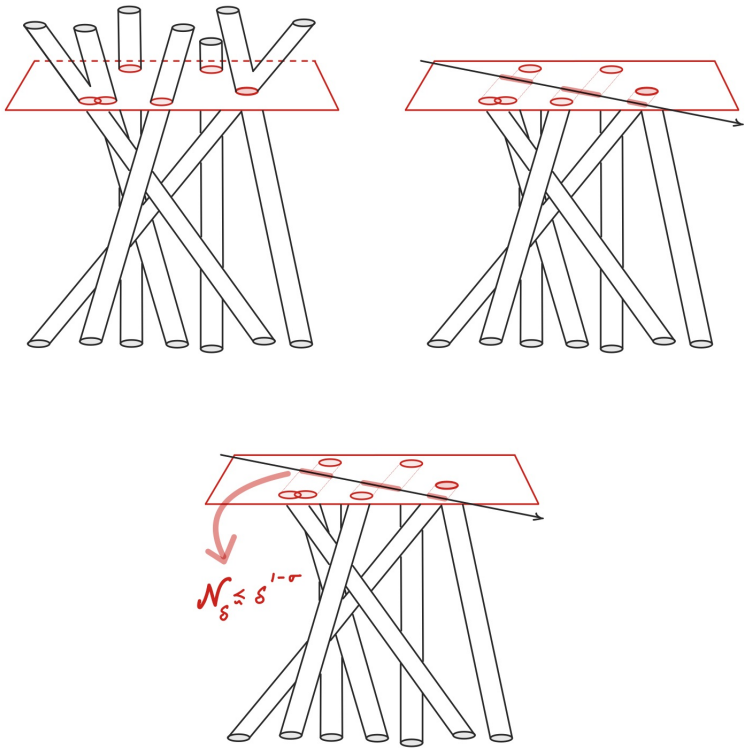
$$|E_{\mathbb{T}}| \lesssim \#\text{balls} \times \#\text{slabs in each ball} \times \text{volume of a slab}$$

which gives

$$|E_{\mathbb{T}}| \lesssim \delta^{\sigma-\varepsilon}\rho^{-3/2} \times \delta^{-\varepsilon}\rho^{\sigma/2-1/2} \times \rho^2 = \delta^{\sigma-2\varepsilon}\rho^{\sigma/2}$$

Note that $\delta \leq \sqrt{\rho}$, which gives $|E_{\mathbb{T}}| \lesssim \delta^{-2\varepsilon}\rho^{\sigma}$. This containment is nearly sharp.

Remark. The second part of the proposition — forces the following structure on each slice of the family, as show below.



As discussed previously, this implies that the projection of each slice along $(1, f(z_0), 0)$ is an ADset, say G, satisfying

$$N(G, \delta) \leq \delta^{-\varepsilon} \left(\frac{1}{\delta}\right)^{1-\sigma} = \delta^{\sigma-1-\varepsilon}$$

So, G can be covered by atmost $\delta^{\sigma-1-\varepsilon}$-many intervals of length δ. If we were to extend these covering intervals along the slice, we would get rectangles with dimensions of the order $1 \times \delta$ which would cover the slice $z = z_0$. So, on each slice, the set $E_{\mathbb{T}}$ is a union of rectangles (called global grains).

Since there are atmost $\delta^{\sigma-1-\varepsilon}$-many rectangles of dimension $1 \times \delta$, by Fubini, $|E_{\mathbb{T}}| \lesssim \delta^{\sigma-\varepsilon}$. Furthermore, $|E_{\mathbb{T}}| \geq M(\varepsilon, \varepsilon, \delta) \gtrsim \delta^{\sigma+\varepsilon}$ for small δ. So, this containment is nearly sharp.

The only information missing is how $f(z_0)$ changes as z_0 varies. We know that it is 1-Lipschitz, but apart from that, we do not have any information about the regularity of the function. In fact, this function turns out to be non-singular C^2. This is perhaps the single most non-trivial step in the proof of the Sticky Kakeya conjecture and uses deep results, that are beyond the scope of this report.

However, we do state the result.

Proposition 4.6. For all $\varepsilon, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$ and an ε-extremal family $(\mathbb{T}, Y)_\delta$ with the following properties.

1. $E_{\mathbb{T}}$ **is a union of local grains with a Lipschitz plane map**:

   There is a 1-Lipschitz plane map $V : E_{\mathbb{T}} \to S^2$, such that for all $p \in \mathbb{R}^3, \rho \in [\delta, 1]$, $V(p) \cdot (B(p, \sqrt{\rho}) \cap E_{\mathbb{T}})$ is an $(\rho, 1-\sigma, \delta^{-\varepsilon})$-ADset.

2. $E_{\mathbb{T}}$ **is a union of global grains with a C^2 slope function**:

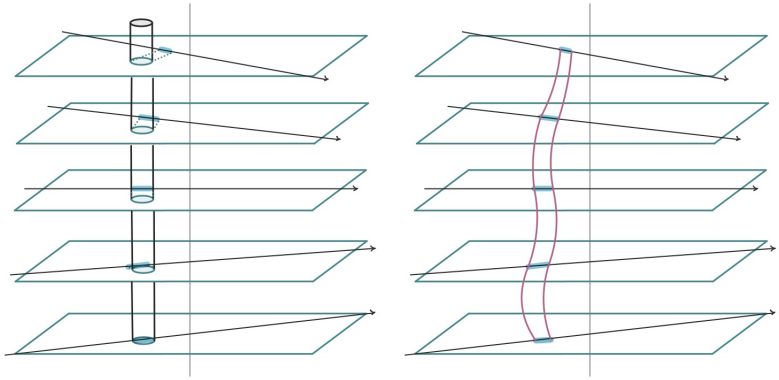
   There is a 1-Lipschitz slope function $f : [-1, 1] \to \mathbb{R}$, such that for all $z \in [-1, 1]$, $(E_{\mathbb{T}} \cap \{z = z_0\}) \cdot (1, f(z_0), 0)$ is an $(\delta, 1-\sigma, \delta^{-\varepsilon})$-ADset.

   Furthermore, for all $z \in [-1, 1]$, $1 \le |f'(z)| \le 2$ and $|f''(z)| \le 1/100$.

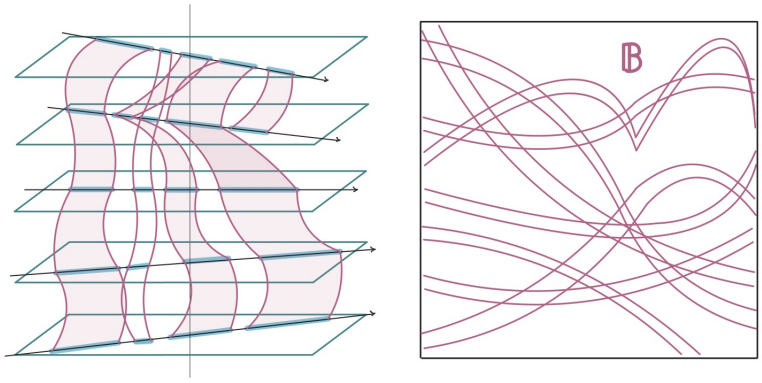
5 Proof of the Sticky Kakeya Conjecture

First, we will reinterpret Proposition — as a statement about twisted projections along the directions given by $(1, f(z_0), 0)$. This effectively translates the problem from 3D to 2D and allows us to use a special case of Wolff's $L^{3/2}$ circular maximal function bound for C^2 curves.

For a δ-tube $T \in \mathbb{T}$, here is how its projection along the directions $(1, f(z_0), 0)$ looks for $z_0 \in [-1, 1]$.



We will take these projection "bands" for each $T \in \mathbb{T}$, and lay them flat onto the square $[-1, 1]^2 \subseteq \mathbb{R}^2$.



To this end, we define.

Definition 5.1 (Twisted Projections)**.** Given $f : [-1,1] \to \mathbb{R}$, the twisted projection is the function $\pi_f : [-1,1]^3 \to \mathbb{R}^2$ defined by

$$\pi_f(x,y,z) := (x + f(z)y, z)$$

So, on a π_f-z plane, the projection of the slice $E_{\mathbb{T}} \cap \{z = z_0\}$ is mapped to the line $z = z_0$. Since each projected slice can be covered by $\delta^{\sigma-1-\varepsilon}$-many intervals of length δ, using Fubini, we have $|\pi_f(E_{\mathbb{T}})| \leq \delta^{\sigma-\varepsilon}$.

Corollary 5.2. For all $\varepsilon > 0, \delta_0 > 0$, there exists $\delta \in (0, \delta_0]$; an ε-extremal collection of δ-tubes $\mathbb{T}$; and a function $f : [-1,1] \to \mathbb{R}$ with $1 \leq |f'(z)| \leq 2$ and $|f''(z)| \leq 1/100$ for all $z \in [-1,1]$, so that

$$\left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^{3/2}(\mathbb{R}^2)} \geq \delta^{-1-\sigma/3+\varepsilon}$$

Proof.

$$\left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^1(\pi_f(E_{\mathbb{T}}))} = \int_{\pi_f(E_{\mathbb{T}})} \left| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right| = \int_{\pi_f(E_{\mathbb{T}})} \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))}$$

$$= \sum_{T \in \mathbb{T}} |\pi_f(Y(T))| \geq \sum_{T \in \mathbb{T}} \frac{|Y(T)|}{\delta} \geq \delta^{-1+\varepsilon}$$

Applying Holder's inequality with conjugates $\frac{1}{3} + \frac{1}{3/2} = 1$

$$\delta^{-1+\varepsilon} \leq \left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^1(\pi_f(E_{\mathbb{T}}))} \leq |\pi_f(E_{\mathbb{T}})|^{1/3} \left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^{3/2}(\mathbb{R}^2)}$$

The conclusion now follows from $|\pi_f(E_{\mathbb{T}})| \leq \delta^{\sigma-\varepsilon}$. $\square$

Definition 5.3 (Cinematic functions)**.** Let $I \subseteq \mathbb{R}$ be a compact interval and $\mathcal{F} \subseteq C^2(I)$. The family $\mathcal{F}$ is said to be a family of cinematic functions with *cinematic constant* K and *doubling constant* D if the following hold.

1. $\mathcal{F}$ has diameter at most K.
2. $\mathcal{F}$ is a doubling metric space, with doubling constant at most D.
3. For all $f, g \in \mathcal{F}$, we have

$$\inf_t \left(|f(t) - g(t)| + |f'(t) - g'(t)| + |f''(t) - g''(t)|\right) \geq K^{-1}\|f - g\|_{C^2(I)}$$

For $f \in C^2(I)$ and $a, b, d \in \mathbb{R}$, define

$$g_{a,b,d}(t) = a + bf(t) + dtf(t)$$

and $M_f = \{g_{a,b,d} : a, b, d, \in [-1, 1]\}$. The following lemma shows that under suitable conditions, M_f is a family of cinematic functions.

Lemma 5.4. Let $f : [0, 1] \to \mathbb{R}$ with $f(0) = 0$, $1 \leq |f'(t)| \leq 2$, and $|f''(t)| \leq 1/100$ for all $t \in [0, 1]$. Then M_f is a family of cinematic functions with constants independent of the choice of f.

The Frostman non-concentration condition: $F \subseteq C^2(I)$ is said to satisfy the Frostman non-concentration condition with $\delta > 0$ if

$$\#(F \cap B) \leq \delta^{-\varepsilon} \left(\frac{r}{\delta}\right) \text{ for all balls } B \subseteq C^2(I) \text{ of diameter } r$$

Fix a family of cinematic functions with constants K and D, say $\mathcal{F} \subseteq C^2(I)$. We state the following theorem without proof.

Theorem 5.5. For all $\varepsilon > 0$, there exists $\delta_0 > 0$ (depending on ε, K, D) such that the following holds for all $\delta \in (0, \delta_0]$.

IF: $F \subseteq \mathcal{F}$ is a δ-separated set that satisfies the Frostman non-concentration condition with δ,

THEN: given f^δ, the δ-neighborhood of the graph of f,

$$\left\| \sum_{f \in \mathcal{F}} \chi_{f^\delta} \right\|_{L^{3/2}} \leq \delta^{-\varepsilon}$$

Proposition 5.6. For all $\varepsilon > 0$, there exists $\delta_0 > 0$ such that the following holds for all $\delta \in (0, \delta_0]$.

IF: $\mathbb{T}$ is a set δ-tubes pointing in δ-separated directions, and f is the slope function as in Proposition 6

THEN:

$$\left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^{3/2}(\mathbb{R}^2)} \leq \delta^{-1-\varepsilon}$$

Comparing Corollary 1 and Proposition 7, we conclude that $\sigma = 0$.

Proof. Without loss of generality, we can assume each $T \in \mathbb{T}$ has a coaxial line along $(c, d, 1) \in \mathbb{R}^3$ with $c, d \in \delta\mathbb{Z} \cap [-1, 1]$ and by δ-separatedness of direction, $(c, d, 1)$'s are distinct for distinct $T \in \mathbb{T}$.

Fix $c \in [-1, 1]$ and let $N = \lceil \delta^{-1} \rceil$. Depending on c, we can choose a set of indices $\mathcal{I} \subset \{-N, \ldots, N\}$ such that for each $i \in \mathcal{I}$, $T_i \in \mathbb{T}$ is a δ-tube with coaxial line $L_i := \left\{ (a_i, b_i, 0) + \mathbb{R}(c, i\delta, 1) : (a_i, b_i) \in [-\frac{1}{3}, \frac{1}{3}]^2 \right\}$.

We observe that $\pi_f(L_i) = \{(t, g_i(t)) : t \in [-1, 1]\}$ where $g_i(t)$ is

$$g_i(t) = g_{a_i, b_i, d_i}(t) + c(t)$$

Also note that $\pi_f(T_i)$ is contained in the 6δ-neighborhood of $\pi_f(L_i)$.

$\mathcal{F} := \{g_{a,b,d}(t) + ct : a, b, d \in [-1, 1]\}$ is a family of cinematic functions by **Lemma 1**, with constants independent of c, f, and N.

The set of functions $F := \{g_i : i \in \mathcal{I}\} \subseteq \mathcal{F}$ is clearly δ-separated. Furthermore, they also satisfy the Frostman non-concentration condition. Thus, we have

$$\left\| \sum_{i \in \mathcal{I}} \chi_{\pi_f(T_i)} \right\|_{L^{3/2}(\mathbb{R}^2)} \leq \delta^{-\varepsilon}$$

Now, we can choose at most δ^{-1}-many $c \in [-1, 1]$ such that their index sets $\mathcal{I}_c$ cover all the tubes $T \in \mathbb{T}$. Applying triangle inequality,

$$\left\| \sum_{T \in \mathbb{T}} \chi_{\pi_f(Y(T))} \right\|_{L^{3/2}(\mathbb{R}^2)} \leq \delta^{-1-\varepsilon}$$

□

6 Conclusion

The path from Besicovitch's 1919 construction to the resolution of the sticky Kakeya conjecture (and eventually the Kakeya conjecture) for $n = 3$, illustrates a recurring patterin in this area of analysis: a classical, purely geometric

obstruction (measure-zero Kakeya sets) forces the introduction of a finer invariant (Hausdorff / Minkowski dimension); proving the resulting conjecture for all Kakeya sets is intractable directly, but becomes tractable once attention is restricted to a rigid, extremal sub-class. For sticky Kakeya sets in $\mathbb{R}^3$, that rigidity takes the concrete form of three structural properties — coarse self-similarity, plaininess, and graininess — each a consequence of the interplay between stickiness hypotheses and the extremality of the exponent σ.

These three properties are together strong enough to feed a argument against the classical planar estimates of Wolff, forcing σ to vanish. The remaining and substantially longer part of Wang & Zahl's resolution of the full $n = 3$ Kakeya conjecture — reducing an arbitrary Kakeya set to an approximately sticky configuration at each scale — lies beyond the scope of this report.

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